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Critical Path Method Exercises

Based on the *PMBOK® Guide Sixth Edition*

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Practice Exercise

Critical Path Method

Exercise 1:

For the following table of information,

1. Draw the network diagram
2. List the network paths
3. Determine the critical path(s)
4. Determine the float for each activity

Activity	Duration	Dependency	Float
Start	0 days	-	
A	5 days	Start	
B	2 days	Start	
C	3 days	A, B	
D	5 days	Start	
E	6 days	Start	
F	4 days	D, E	
G	2 days	C, F	
H	5 days	G	
I	7 days	G	
J	3 days	H	
Finish	0 days	I, J	

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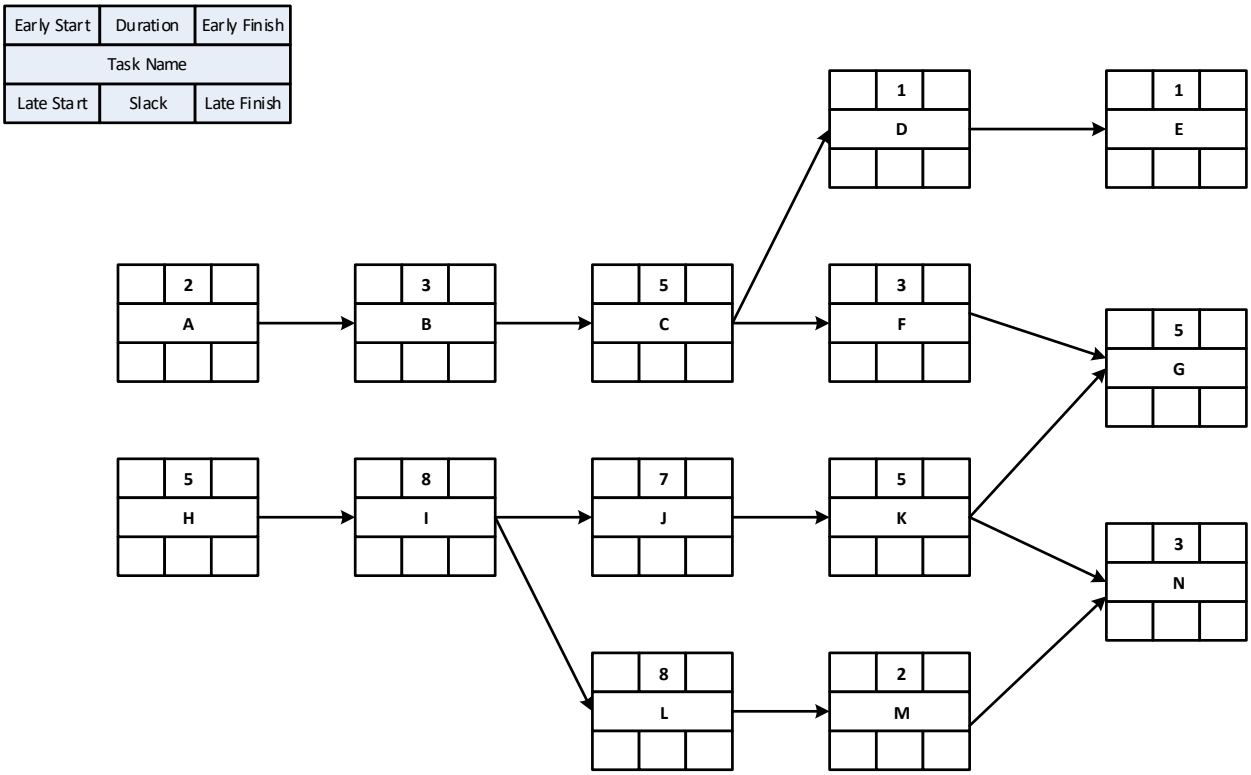
Exercise 2:



1. Perform a Forward Pass and a Backward Pass of the network diagram above

2. Which Activities has the same Early Finish as they have Late Finish?

Note: The project starts on Day "0"

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Exercise 3:

For the following table of information,

1. Draw the network diagram
2. List the network paths
3. Determine the critical path(s)
4. Determine the float for each activity

Activity	Duration (Days)	Dependency	Float
Start	0	-	
A	8	Start	
B	5	A	
C	6	B, H	
D	8	C	
E	9	D, F	
F	3	C, I	
G	7	Start	
H	8	G	
I	9	H	
J	5	I	
K	2	J	
L	3	J	
Finish	0		

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Critical Path Method Exercise *Solutions*

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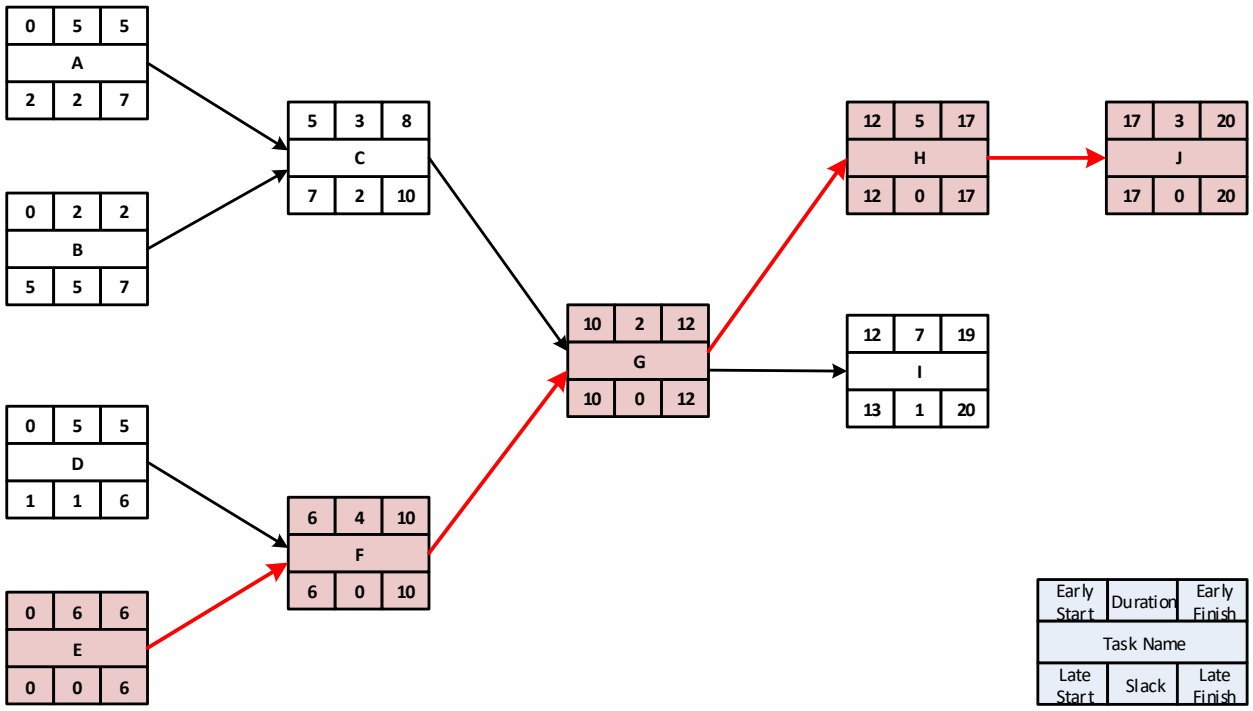
Exercise 1 Solution:

1.



2. Network paths:

1. **A -> C -> G -> H -> J**
2. **A -> C -> G -> I**
3. **B -> C -> G -> H -> J**
4. **B -> C -> G -> I**
5. **D -> F -> G -> H -> J**
6. **D -> F -> G -> I**
7. **E -> F -> G -> H -> J**
8. **E -> F -> G -> I**

3. Critical path: **E -> F -> G -> H -> J (shown in red)**

4. Activity floats shown in the diagram

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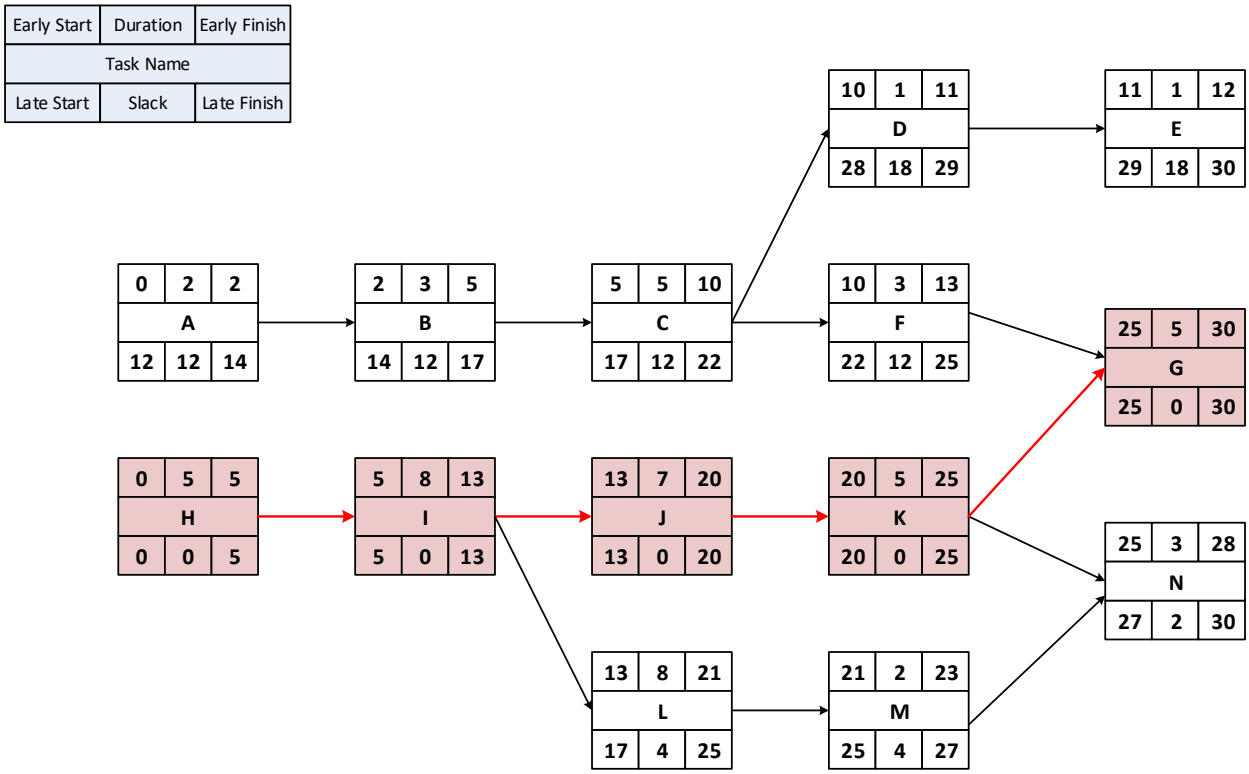
Exercise 2 Solution:

1.



2. Any activity on the critical path will have Late Finish equal to its Early Finish

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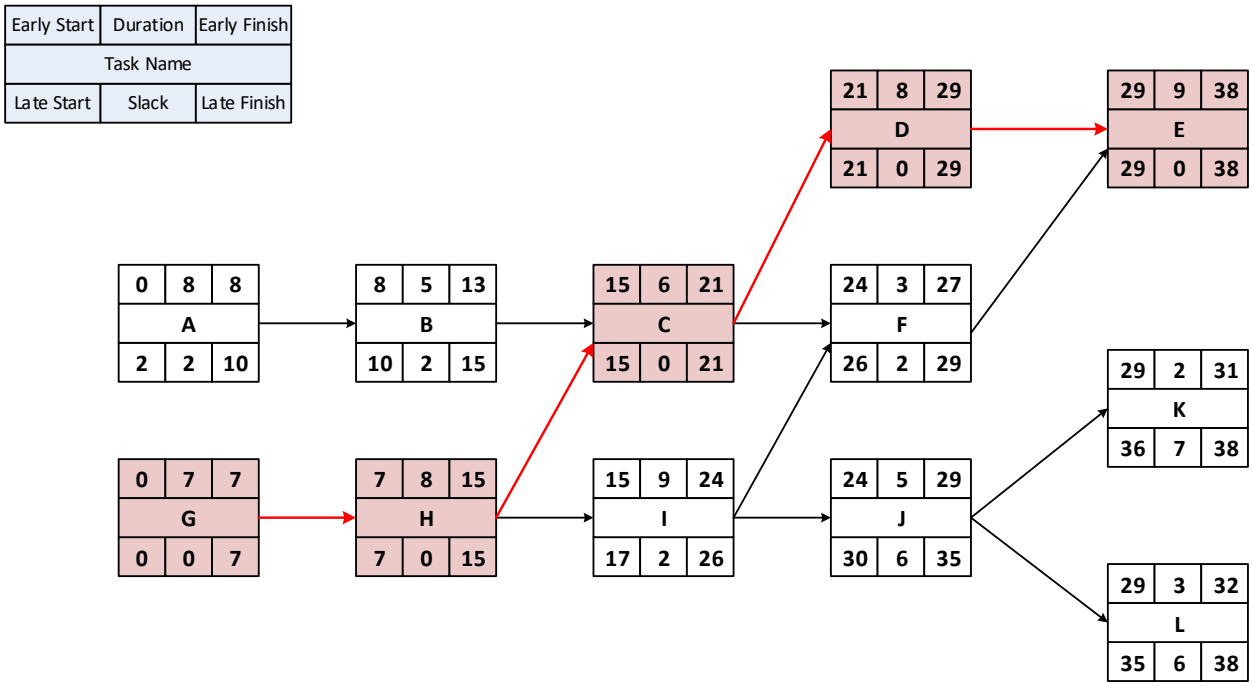
Exercise 3 Solution:

1.



2. Network paths:

1. **A -> B -> C -> D -> E**
2. **A -> B -> C -> F -> E**
3. **G -> H -> I -> J -> K**
4. **G -> H -> I -> J -> L**
5. **G -> H -> C -> D -> E**
6. **G -> H -> C -> F -> E**
7. **G -> H -> I -> F -> E**

3. Critical path: **G -> H -> C -> D -> E (shown in red)**

4. Activity Floats shown in the network diagram

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